University of Saskatchewan
MATH 133.4: Engineering Mathematics I
Fall 2025-2026
Course Outline

1 Treaty acknowledgement

As we gather here today, we acknowledge that the Saskatoon campus of the University of Saskatchewan is on Treaty Six Territory and the Homeland of the Métis. We pay our respect to the First Nation and Métis ancestors of this place and reaffirm our relationship with one another. We recognize that in the course of your studies you will spend time learning in other traditional territories and Métis homelands. We wish you safe, productive and respectful encounters in these places.

2 Instructional team and contact information

Instructor:	Dr. Gary Au (au@math.usask.ca)
Course Coordinator:	Zoe Mao (zoe.mao@usask.ca)
Lab coordinators:	Manuela Golban (golban@math.usask.ca)
	Amos Lee (lee@math.usask.ca)

In-person lectures are held in ARTS 143 on Mondays, Wednesdays, and Fridays at

- 9:30AM - 10:50AM for Blocks 01, 02, 05, 06, 09, and 10;
- 1:30PM - 2:50PM for Blocks 03, 04, 07, 08, 11, and 12.

3 Course description and prerequisites (from the Catalogue)

Course description: An introduction to foundational concepts and tools in calculus, linear algebra, and statistics that are essential to engineering. Topics include basic integration techniques, limits and continuity, derivatives and their applications, matrix operations and linear transformations, linear regression, and graphing data on various scales.

Prerequisites: Mathematics B30 and C30, or Pre-Calculus 30, or MATH 102.

Note: Enrolment is restricted to students in the College of Engineering. A successful completion of this course satisfies the Quantitative Reasoning core curriculum requirement of the College of Arts and Science.

4 Learning outcomes

The following are the ten groups of tasks and concepts that students will be expected to master by the end of the course:

CLO1 Module 1

RLO1 **Basic Differentiation and Integration.** Be able to interpret the derivative as the slope of a tangent line to a curve, and understand the definite integral as the net area under a curve over an interval. Be able to apply the power rule for differentiation and integration.

RLO2 **Limits and Continuity.** Understand the concept of a limit and its basic properties. Determine whether a limit exists, and compute it if it does. Understand the notion of continuity, and distinguish between removable and non-removable discontinuities. Find vertical and horizontal asymptotes of functions.

RLO3 **Computing Derivatives I.** Understand the definition of the derivative as the limit of average rate of change approaching instantaneous rate of change. Understand the common interpretations of the derivative as slopes of tangent lines, instantaneous rates of change of a function, and in particular the velocity of an object in rectilinear motion. Be able to compute derivatives using the product, quotient, and chain rules. and compute derivatives of exponential functions.

RLO4 **Computing Derivatives II.** Be able to evaluate derivatives of trigonometric and inverse trigonometric functions. Apply implicit differentiation to evaluate derivatives for implicitly defined functions. Differentiate logarithmic functions, and apply logarithmic differentiation.

RLO5 **Matrices and Linear Systems I.** Be familiar with common terminology involving systems of linear equations, including augmented matrices, elementary row operations, row-echelon form (REF) and reduced row-echelon form (RREF), and Gaussian elimination. Utilize Gaussian elimination to solve systems of linear equations, and be able to determine when a system is consistent or inconsistent. Perform basic matrix operations (addition, scalar multiplication and matrix multiplication) and solve linear systems (up to 3-by-3) on paper. Find the transpose and (when applicable) the inverse of a given matrix.

CLO2 Module 2

RLO6 **Matrices and Linear Systems II.** Understand the definition of a linear transformation and be able to find the standard matrix for a given transformation. Be able to perform rotation, reflection, and stretch/compression linear transformations in $\mathbb{R}^2$. Calculate the determinant of a matrix using cofactor expansion and understand the basic properties of the determinant, including its relation to the area of a parallelogram formed by two vectors, as well as invertibility of matrices. Be able to find the characteristic polynomial, eigenvalues, and eigenvectors of a matrix.

RLO7 **Summary Statistics and Linear Regression.** Be able to compute basic summary statistics (e.g. mean, median, mode, sample variance, and sample standard deviation) of a given set of data. Understand the concept of percentiles and quartiles, and construct box-and-whisker plots of a given set of data. Find the line of best fit for a given set of discrete data using the method of least squares. Obtain estimates using interpolation and extrapolation, and understand the limitations of these estimates. Articulate what we can and cannot conclude about the underlying set of data given its summary statistics. Use a computational tool (such as Matlab or Microsoft Excel) to perform basic statistical computations.

RLO8 **Applications of Derivatives I.** Understand the basic connections between the first two derivatives with the extrema and concavity of a given function. Be able to find absolute extrema of a function on a closed interval. Solve optimization problems of one variable. Solve problems involving quantities with related rates of change.

RLO9 **Applications of Derivatives II.** Calculate the linearizations of simple functions and use it for approximations. Find approximate solutions to equations using Newton's method. Use L'Hospital's rule to evaluate limits of indeterminate forms when appropriate. Be able to recognize a differential equation.

RLO10 **Graphing Data.** Recognize rectilinear, semi-log, and log-log grids, and be able to plot a given set of points on them. Understand the key elements of common graphing procedures in engineering. Determine representative equations from rectilinear, semi-log, and log-log plots, and verify the equation against observed values.

5 Required course materials

To successfully complete this course, you'll need:

1. A Möbius subscription (which you should already have for having completed the Jumpstarts);
2. Regular access to internet and a laptop/mobile device to perform various tasks (e.g. submit assignments) and stay connected with the course.

In particular, there is no required textbook for this course. All course materials will be provided throughout the term.

6 Course organization and schedule

Table 1 provides a detailed schedule of our course.

Week/Date	Activities and Events	Course Topic
1 (Sep. 1-5)	(Mon.) No classes — Labour day	RLO1: Basic Differentiation and Integration
2 (Sep. 8-12)	(Mon.) Assignments 1A and 1B released (Fri.) Assignments 2A and 2B released	RLO2: Limits and Continuity
3 (Sep. 15-19)	(Mon.) Assignments 2A and 2B released (Wed.) Last day to make registration changes with 100% tuition credit (Fri.) Assignments 1A and 1B due	RLO3: Computing Derivatives I
4 (Sep. 22-26)	(Mon.) Assignments 3A and 3B released (Thu.) Last day to withdraw with 75% tuition credit (Fri.) Assignments 2A and 2B due	RLO4: Computing Derivatives II
5 (Sep. 29 - Oct. 3)	(Mon.) No classes — National Day of Truth and Reconciliation (Mon.) Assignments 4A and 4B released (Thu.) Last day to withdraw with 50% tuition credit (Fri.) Assignments 3A and 3B due	RLO5: Matrices and Linear Systems I
6 (Oct. 6-10)	(Mon.) Assignments 5A and 5B released (Fri.) Assignments 4A and 4B due	RLO5: Matrices and Linear Systems I (Continued)
7 (Oct. 13-17)	(Mon.) No classes — Thanksgiving (Wed.) No classes — GE 132 (Fri.) Assignments 5A and 5B due	RLO6: Matrices and Linear Systems II
8 (Oct. 20-24)	(Mon.) Assignments 6A and 6B released (Wed.) No classes — GE 132 (Wed.) Module Test 1	RLO7: Summary Statistics and Linear Regression
9 (Oct. 27-31)	(Mon.) Assignments 7A and 7B released (Fri.) Assignments 6A and 6B due	RLO8: Applications of Derivatives I
10 (Nov. 3-7)	(Mon.) Assignments 8A and 8B released (Fri.) Assignments 7A and 7B due	RLO9: Applications of Derivatives II
11 (Nov. 10-14)	No classes — Fall Break	
12 (Nov. 17-21)	(Mon.) Assignments 9A and 9B released (Mon.) Assignments 10A and 10B released (Fri.) Assignments 8A and 8B due (Sat.) Top-Up Module Test 1	RLO10: Graphing Data
13 (Nov. 24-28)	(Fri.) Assignments 9A and 9B due (Fri.) Assignments 10A and 10B due	(Additional topics and/or Review)
14 (Dec. 1-5)	(Wed.) Module Test 2 (Fri.) Last day to withdraw with no tuition credit	(Additional topics and/or Review)
16 (Dec. 15-19)	(Fri.) Top-Up Module Test 2	

Table 1: A detailed schedule for MATH 133

7 Assessment and evaluation

Important Note: While we are implementing some elements from competency-based assessments (CBA) into our course, MATH 133 uses a conventional grading scheme that has significant differences compared to the fully-CBA grading schemes used in many of your GE courses. Make sure you read the following carefully to understand how your grades will be computed.

7.1 Grading scheme

Your final grade (out of 100) of this course will be calculated based on the scheme shown in Table 2.

Module	Component	Points
1	Assignments	8
	Module Test (Type B)	30
	Module Test (Type C)	12
2	Assignments	8
	Module Test (Type B)	30
	Module Test (Type C)	12
Total		100

Table 2: Grade breakdown for MATH 133

To pass the course, a student must achieve at least 70% on the Type B portion of the Module Test of each module. In particular, there's no minimum requirements on the Assignments or Module Test (Type C) credits, nor do they contribute towards passing the course.

A few more notes on grade computation:

- All assignments in a module are weighted equally in computing the Assignment component score.
- If a student scores higher on Module Test (Type B) than an earlier assignment in the same module, the earlier and lower assignment score is replaced by the later and higher score. For example, if a student scores 80% on the Type B portion of Module Test 2, all of their assignments in Module 2 that were lower than 80% "become" 80% in the computation of their final course grade. This CBA-like policy is aimed to reward students who show mastery of the material on a module test.
- If a student scores lower than 70% on the Type B portion of the initial module test and then scores at least 70% on the corresponding top-up module test, their Module Test (Type B) score of that module becomes exactly 70%, and they pass the module. And if a student scores lower than 70% on the Type B portion of both the initial module test and the top-up, they would receive a failing score for the module (and hence the course).

If a student meets all of the conditions for passing the course, their final course grade would be that computed from Table 2. On the other hand, if a student does not meet the passing requirements above, their final course grade would be either 49 or their computed grade from Table 2, whichever is lower.

7.2 Course assessments

7.2.1 Summer Jumpstarts

The MATH1 and MATH2 Jumpstarts were designed to help ensure that you have the necessary mathematics background to succeed in your first-year university courses. They have been available since July, and are due by 11:59PM on Friday September 5th. Students must complete the two Jumpstarts (by passing every unit therein) to receive a grade for Assignment 1A of this course.

7.2.2 Möbius Assignments

Each of the ten RLOs comes with an Assignment A and an Assignment B. They are both electronic assignments hosted in Möbius.

- Assignment A covers basic concepts of the unit and aims to lay the foundation for more challenging material. It has no limit on the number of attempts allowed. Each attempt will be instantly graded, and the highest score will be kept. You may get different questions for each attempt.

  For each Assignment A, there is a passing threshold (of around 90%), which will be indicated in Möbius. One must pass the Assignment A before the corresponding Assignment B is available for attempt. (That is, if you don't complete an Assignment A by the due date, you'd also receive a grade of zero on the Assignment B.)

- Assignment B consists of a variety of problems that can be slightly more challenging than those on Assignment A and aim to develop a deeper level of understanding. It allows a maximum of 3 attempts. Each attempt will be instantly graded, and the highest score will be kept. You may get different questions for each attempt.

7.2.3 Module Tests

There will be two Module Tests:

- Module Test 1 (covers RLOs 1 to 5) is held on Wednesday Oct. 22nd, 6:00-8:30PM;
- Module Test 2 (covers RLOs 6 to 10) is held on Wednesday Dec. 3rd, 6:00-8:30PM.

Each module test has a Type B portion that consists of more standard multiple-choice and full-solution problems, as well as a Type C portion that presents more challenging problems. More details about the module tests will be announced as they approach.

7.2.4 Top-Up Module Tests

If a student scores **at least 50% and lower than 70%** on the Type B portion of the initial module test, they are eligible to write module's top-up module test for a chance to pass the module. The top-up module test only consists of Type B problems.

- Top-Up Module Test 1 is held on Saturday Nov. 22nd, 8:30-11:00AM;
- Top-Up Module Test 2 is held on Friday Dec. 19th, 9:00-11:30AM.

More details about the top-up module tests will be announced as they approach.

7.3 Make-up policy

Extensions or exemptions will not be granted for Möbius assignments. Under our grading scheme, the Module Test (Type B) grade replaces all lower assignment grades in the module. Thus, if you have to miss a formative assessment for any reason, you would have the opportunity to make that up on the corresponding module test.

If you do have to miss a module test for a valid reason, you should e-mail our course coordinator to explain your situation at least five days before the Module Test day, or no later than 24 hours after the start of the module test for medical emergencies. If you are granted an exemption for a module test, you will be permitted to write a make-up module test on the day of the corresponding top-up module test. Note that students who write a make-up module test will not have a further top-up opportunity for that module.

7.4 Regrading policy

If you feel that you have been graded incorrectly on a Möbius assignment, you should post your concern on our Canvas discussion board. We'd respond there, and this way we keep all the helpful information in one place.

And if you feel that you have been unfairly graded on a module test, you should first consult with and carefully verify your work against the posted solutions. If the concern remains, you should complete a re-grade request form (available on Canvas) and e-mail it to our lab coordinator Manuela Golban within 48 hours of the graded assessment being available.

7.5 A general rubric

Throughout the course, students' written work (on module tests) will be graded using the following general rubric:

- 10/10 **Mastery** — Submitted work is complete, correct, and well communicated. Mastery of the relevant materials are evident.
- 9/10 **Developing Mastery** — Submitted work is complete, correct, and well communicated barring a small error that does not trivialize the underlying task. Mastery of the relevant materials are evident despite the error.
- 7/10 **Competence** — Understanding of the relevant materials is evident, but the submitted work requires a non-trivial revision and/or expansion.
- 5/10 **Developing Competence** — Partial understanding of the relevant materials is evident, but significant gaps remain. Submitted work requires substantial revision and/or expansion.
- 3/10 **Not Yet Competent** — Submitted work is fragmented, with significant omissions and/or flaws. Not enough information to determine if there's understanding of the material.
- 0/10 **No Evidence of Competence** — Submitted work is devoid of substance, or no work was submitted at all.

8 Help channels

8.1 Lab tutorials

There are a number of lab sections associated with MATH 133. See your program schedule for when and where your lab tutorials take place. Do note the time of your lab tutorial can vary from week to week.

During the lab period, the Lab Instructor will be available to answer student questions, as well as to provide further examples from the lecture topics. You should attend the lab section for which you are registered.

8.2 First-year engineering help sessions

The first-year engineering help sessions take place in ENG 2C02. A teaching assistant will be available on Wednesdays and Fridays from 4:30 PM to 5:30 PM to assist with MATH 133.

8.3 Canvas discussion board

We will use the discussion board in our Canvas course page. All questions related to the content and logistics of our course should be posted there. We will monitor this board regularly.

8.4 E-mailing your instructor

For matters that are personal in nature, you are welcomed to e-mail us individually. If you don't know who to specifically write to, you should e-mail our course coordinator Zoe Mao (zoe.mao@usask.ca) and cc your instructor. Please do note if a question is deemed to be general in nature, you'd be directed to post on the Canvas discussion board instead, so we can keep all the helpful information in one place.

If you do e-mail us, make sure you include MATH 133 in the subject of your message, and e-mail from the usask.ca domain. (E-mails from external domain have a higher chance of being auto-sorted into the spam folder.)

9 Some Notes and Thoughts on Academic Integrity

9.1 So what IS cheating?

A reasonable definition of academic cheating is "taking others' work and passing it off as your own." Clear-cut examples of cheating include submitting plagiarized assignments or module test solutions prepared by someone else, such as a peer, a private tutor, or a service like Chegg, Symbolab, or generative AI.

However, if the "others' work" consists of course materials provided by the instructor or ideas shared in public discussions by your peers on the Canvas discussion board, then using those resources is entirely acceptable. They are intended to support students in their learning.

Some situations are less straightforward. For instance, if you have a friend who has taken or is currently taking the course and is helping you with the material, the key consideration is whether you are using that assistance to genuinely learn the content or merely to complete your homework. After receiving help, ask yourself if you can explain the concepts in your own words and if you would be able to independently solve a similar problem three weeks later. If the answer is yes, it indicates that meaningful learning has occurred.

If you are uncertain whether a particular form of assistance constitutes cheating, we strongly encourage you to discuss it with us first. This way, we can ensure you receive the help you need through legitimate means.

9.2 Why cheating is bad

Simply put, now is the time to cultivate the good habits you will need as you take on greater responsibilities and face more significant challenges in the future. Cheating now robs you of the opportunity to learn and grow in a relatively safe environment.

While a cheater may avoid immediate consequences, such as losing grades or failing a course, the long-term costs can be even greater. As the gap between your perceived and actual abilities widens, you may find yourself increasingly reliant on dishonest practices, making it harder to succeed through legitimate means. When you are eventually exposed, the repercussions could be far more severe than simply losing a few points in a course.

We genuinely want you to succeed and are committed to facilitating your learning. If you feel overwhelmed or unsure about how to proceed in the course, please reach out to us and share your situation.

To the majority of students who are working through the course honestly and may feel frustrated by the possibility — or knowledge — of peers cheating to get ahead, we understand your concerns. We urge you not to fall into that trap for all the reasons mentioned above. Trust that doing the right thing today will serve you well in the future. We promise to do everything within our power to maintain a healthy and fair learning environment.

9.3 Academic integrity policies at the U of S

The University of Saskatchewan is committed to the highest standards of academic integrity.

https://academic-integrity.usask.ca/

Students are urged to read the Regulations on Academic Misconduct and to avoid any behaviours that could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence.

For help developing the skills for meeting academic integrity expectations, see:

https://academic-integrity.usask.ca/students.php

Students are encouraged to ask their instructors for clarification on academic integrity requirements.

All students are encouraged to be aware of the rules for courses set out in the Academic Courses Policy on Class Delivery, Examinations, and Assessment of Student Learning.

10 Academic Courses Policy

The USask Academic Courses Policy contains the requirements for course delivery, examinations, and other forms of student assessment. You can view the policy at

https://policies.usask.ca/policies/academic-affairs/academic-courses.php.

You can also read information on the following policies and procedures at the links below:

- https://governance.usask.ca/student-conduct-appeals/academic-misconduct.php
- https://governance.usask.ca/student-conduct-appeals/non-academic-misconduct.php
- https://governance.usask.ca/student-conduct-appeals/appeals-in-academic-matters.php

11 Copyright

Course material created by your professors and instructors is their intellectual property and cannot be shared without written permission. This includes exams, PowerPoint/PDF lecture slides and other course notes. If materials are designated as open education resources (with a creative commons license) you can share and/or use them in alignment with the CC license. Other copyright-protected materials created by textbook publishers and authors may be provided to you based on license terms and educational exceptions in the Canadian Copyright Act.

You are responsible for ensuring that any copying or distribution of materials that you engage in is permitted by the University's "Use of Materials Protected By Copyright" Policy. For example, posting others' copyright-protected materials on the open internet is not permitted by this policy unless you have copyright permission or a license to do so. For more copyright information, please visit

https://library.usask.ca/copyright/students/index.php

or contact the University Copyright Coordinator at copyright.coordinator@usask.ca or 306-966-8817.

12 Access and Equity Services (AES) for Students

Access and Equity Services (AES) is available to provide support to students who require accommodations due to disability, family status, and religious observances.

Students who have disabilities (learning, medical, physical, or mental health) are strongly encouraged to register with Access and Equity Services (AES) if they have not already done so. Students who suspect they may have disabilities should contact AES for advice and referrals at any time. Those students who are registered with AES with mental health disabilities and who anticipate that they may have responses to certain course materials or topics, should discuss course content with their instructors prior to course add / drop dates.

Students who require accommodations for pregnancy or substantial parental/family duties should contact AES to discuss their situations and potentially register with that office.

Students who require accommodations due to religious practices that prohibit the writing of exams on religious holidays should contact AES to self-declare and determine which accommodations are appropriate. In general, students who are unable to write an exam due to a religious conflict do not register with AES but instead submit an exam conflict form through their PAWS account to arrange accommodations.

Any student registered with AES, as well as those who require accommodations on religious grounds, may request alternative arrangements for mid-term and final examinations by submitting a request to AES by the stated deadlines. Instructors shall provide the examinations for students who are being accommodated by the deadlines established by AES.

For more information or advice, visit

https://students.usask.ca/health/centres/access-equity-services.php,

or contact AES at 306-966-7273 (Voice/TTY 1-306-966-7276) or email aes@usask.ca.

13 Student Supports

13.1 Academic Help — University Library

Visit the University Library and Learning Hub to find supports for undergraduate and graduate students with first-year experience, study skills, learning strategies, research, writing, math and statistics. Students can attend workshops, access online resources and research guides, book 1-1 appointments or hire a subject tutor through the USask Tutoring Network

Connect with library staff through the AskUs chat service or visit various library locations on campus.

Enrolled in an online course? Explore the Online Learning Readiness Tutorial.

13.2 Teaching, Learning and Student Experience

Teaching, Learning and Student Experience (TLSE) provides developmental and support services and programs to students and the university community. For more information, see the students' website http://students.usask.ca.

13.3 College Supports

Students in Arts & Science are encouraged to contact the Undergraduate Student Office and/or the Trish Monture Centre for Success with any questions on how to choose a major; understand program requirements; choose courses; develop strategies to improve grades; understand university policies and procedures; overcome personal barriers; initiate pre-career inquiries; and identify career planning resources. Contact information is available at

http://artsandscience.usask.ca/undergraduate/advising/.

13.4 Financial Support

Any student who faces unexpected challenges securing their food or housing and believes this may affect their performance in the course is urged to contact Student Central https://students.usask.ca/student-central.php.

13.5 Gordon Oakes Red Bear Student Centre

The Gordon Oakes Red Bear Student Centre) is dedicated to supporting Indigenous student academic and personal success. The Centre offers personal, social, cultural and some academic supports to Métis, First Nations, and Inuit students. The Centre is an intercultural gathering space that brings Indigenous and non-Indigenous students together to learn from, with and about one another in a respectful, inclusive, and safe environment. Visit https://students.usask.ca/indigenous/index.php or students are encouraged to visit the ASC's website https://students.usask.ca/indigenous/gorbsc.php

13.6 International Student and Study Abroad Centre

The International Student and Study Abroad Centre (ISSAC) supports student success and facilitates international education experiences at USask and abroad. ISSAC is here to assist all international undergraduate, graduate, exchange, and English as a Second Language students in their transition to the University of Saskatchewan and to life in Canada. ISSAC offers advising and support on matters that affect international students and their families and on matters related to studying abroad as University of Saskatchewan students. Visit https://students.usask.ca/international/issac.php for more information.